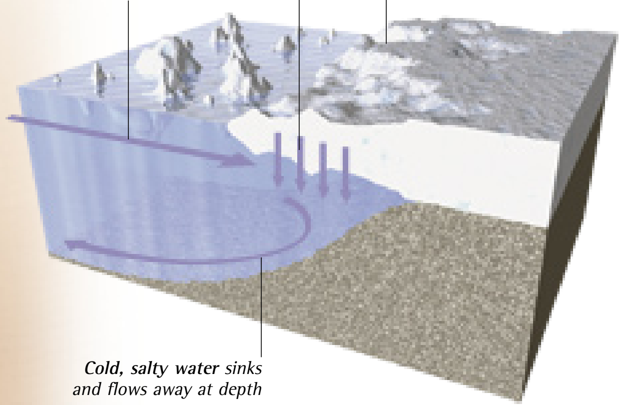
DEEP-WATER CURRENTS

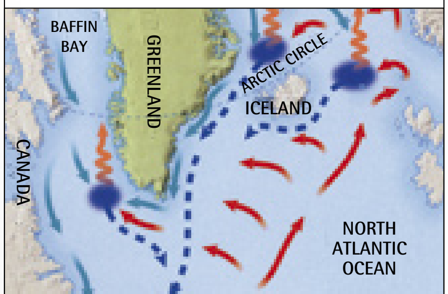
The surface currents that swirl around the oceans are linked to a network of slow-moving deep-water currents that carry ocean water all around the globe. Surface currents are driven by winds, but the forces that drive deep-water currents are more complex. The main mechanism is a change of density, caused by cooling and an increase in saltiness. This makes the water heavier, so it sinks toward the ocean floor. It then flows beneath warmer water, often very slowly, and gradually mixes with it until it returns to the surface. Eventually, this carries every gallon of ocean water around the globe, but the journey can take thousands of years.



▲ SINKING CURRENTS
Most of the sinking water that drives deep-water currents originates in the polar oceans, where floating ice and freezing at the surface make the ocean water colder, denser, and heavier. Salt expelled from ice also makes the water more salty, increasing its density. Due to the twin roles of temperature and saltiness, the mechanism is called the thermohaline circulation, after the Greek words for heat and salt.

◄ ANTARCTIC WATER
The coldest deep-ocean current flows from the Antarctic Weddell Sea, where the thermohaline process operates under the vast Ronne Ice Shelf as well as beneath the drifting pack ice. It creates a current called the Antarctic Bottom Water, which flows eastward across the floor of the Southern Ocean. It merges with a similar cold deep-water current, which is generated in the Ross Sea on the other side of Antarctica.

NORTH ATLANTIC DEEP WATER



KEY
- Cold deep-water current
- Warm surface currents
- Loss of heat
- Cold surface current
- Sinking zone

Most of the cold bottom water created by ice in the far north is retained by rocky barriers around the Arctic Ocean floor. But when warm, salty Atlantic water flowing north meets cold currents, it cools and sinks below the less salty Arctic water. The three main sinking zones drive a current called the North Atlantic Deep Water.